
Verified Mechanisms

Formally verified autoresearch
for theoretical mech interp

Karthik Viswanathan

MATS 9.1

Automated Alignment Research Seminar, September 2026

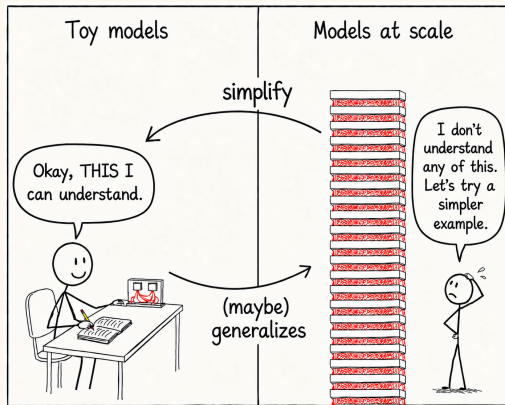


`verifiedmechanisms.ai`



About me

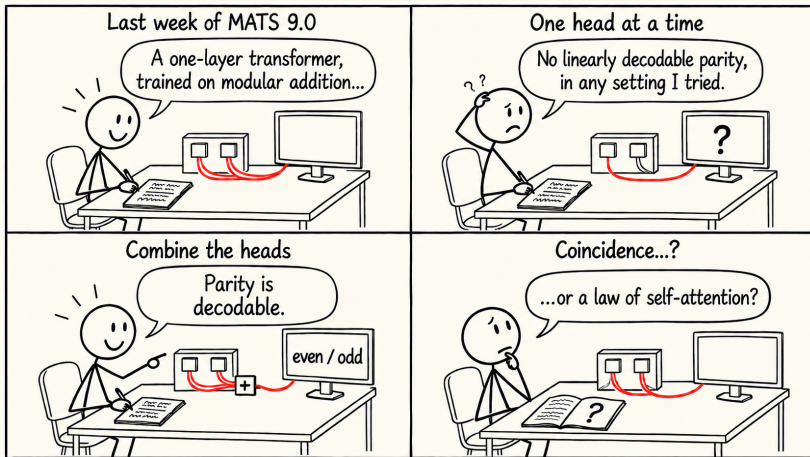
- 1 Now: Simplex (MATS 9.1)**
How does post-training reshape the internal mechanisms? ft. modular addition
- 2 Before: Geodesic (MARS 3.0)**
A transformer architecture alteration to incentivise externalised reasoning
- 3 Research taste**
I end up working on the full range of models, from toy examples to models at scale.



To introduce Verified Mechanisms,
let me begin with a short story.

*A minimal proof of concept showing how formally verified
autoresearch can help mech interp researchers.*

The origin story



In search of a “law of self-attention”



blog post

“Can an attention head compute XOR?”

1

Ask frontier models

While out on a walk, I posed the question to Opus 4.7 and GPT-5.4.

2

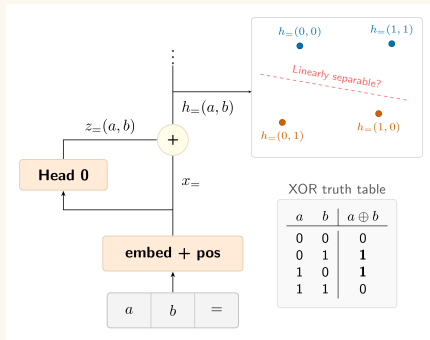
One hour later

Both returned (fairly involved) answers.¹

3

Formalise the proof

I asked Opus to formalise GPT’s argument in Lean 4, and the proof checked out!



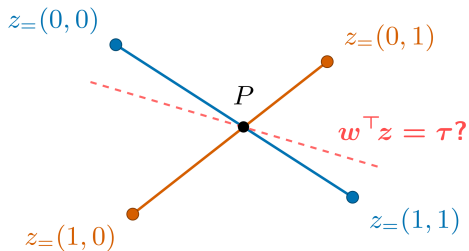
Linearly separable into XOR classes?

¹Opus’ argument even contained a couple of hallucinated steps.

Outcome: first law of self-attention



blog post



Attention outputs $z=(a,b)$ for the four inputs:
blue for even parity, orange for odd parity.

One head cannot compute XOR.

ft. a surprisingly elegant proof by GPT-5.4,
formalised by Opus 4.7.

After a single attention-head update, the
line segment joining the even-parity
outputs always intersects the segment
joining the odd-parity outputs, ruling out
their linear separability.

Questions are easy, proofs are hard

At least, they used to be.

What I wanted to know

**“Can an attention
head compute
XOR?”**

What I would need to analyse

$$h_{=}(a, b) = x_{=} + z_{=}(a, b)$$

$$z_{=}(a, b) = \frac{\sum_{j \in \{a, b, =\}} \exp(x_{=}^{\top} W_Q^{\top} W_K x_j) W_V x_j}{\sum_{j \in \{a, b, =\}} \exp(x_{=}^{\top} W_Q^{\top} W_K x_j)}$$

*The cost of pursuing hard mathematical questions is lower now,
thanks to formally verified autoresearch.*

Slightly scaling up: more laws of self-attention



GitHub

Longer runs: GPT-5.5 conjectures and proves, Claude 4.8 formalises in Lean 4.

The pilot problem

How many heads does it take to represent a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$?

$$H^*(f) = \min\{H : f \text{ is representable by an } H\text{-head model}\}.$$

Theorem (symmetric functions)

A closed form for $H^*(f)$ for every symmetric Boolean function, $H^*(f) = C(f)$.²

Corollary: $H^*(\text{XOR}_n) = n$.

²The proof is left as an exercise for the reader and their LLM.

Zooming out.

From pilot question to research programme

What is Verified Mechanisms?

Formally verified autoresearch for theoretical mech interp

1 Goal

We aim to produce foundational results in mechanistic interpretability by leveraging the recent advances in the mathematical abilities of frontier models.

2 Verification

To make sure that the theorems provided by the frontier models do not contain logical flaws, we formally verify them in Lean 4.



`verifiedmechanisms.ai`

What is Verified Mechanisms?

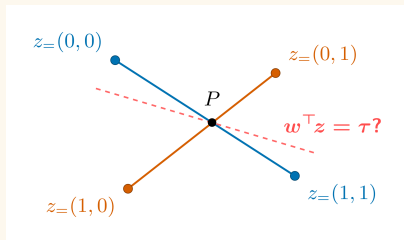
Formally verified autoresearch for theoretical mech interp

1 Goal

We aim to produce foundational results in mechanistic interpretability by leveraging the recent advances in the mathematical abilities of frontier models.

2 Verification

To make sure that the theorems provided by the frontier models do not contain logical flaws, we formally verify them in Lean 4.



First law of self-attention

Theory of impact

Why should we do theoretical mech interp?

Precedent

Foundational theory has shaped mechanistic interpretability before.

Elhage et al. (2021) worked out the mathematics of attention from first principles, and many later mechanistic interpretability ideas (for example, induction heads, transformer circuits) build on that framework.

Our contribution

Our aim is more modest than Elhage et al. but similar in spirit.

A growing library of machine-checked theorems that applied researchers can use as reference points, and a workflow for producing them faster than hand-derivation alone.

Why formally verified autoresearch for theoretical mech interp?

Need for autoresearch

Foundational theory is hard to scale with human effort alone.

Hand-derivation is slow and expertise-intensive, and frontier models can now carry much of this load.

Need for formal verification

Unverified autoresearch can fail quietly.

Long derivations by frontier models can hide subtle errors that human review could miss. Lean checks every step and rejects invalid proofs.

More concretely, how we hope to contribute

- 1 Stronger fundamentals for toy-model research**
Foundational theory of transformer components can provide a principled basis for interpreting and generalising empirical findings from toy models.
- 2 Foundationally grounded probes for models at scale**
Mathematical abstractions for tracking features through a transformer could inform theoretically grounded feature-decomposition tools.
- 3 Infrastructure for other researchers**
A multi-agent framework to prove and verify questions about transformer components.

How are we going to do formally verified autoresearch?

A quick look at the components of the system design.³

³Fair warning: the next couple of slides may sound a little like a capabilities talk.

Scaling up autoresearch is not straightforward



autoresearch branch

Good runs produce useful mathematics. Bad runs produce AI slop.

1

The long run

It produced around 200 lemmas, but only a handful were useful.

2

Burning tokens

The run spent many hours pursuing minor subproblems while spawning more subagents.

3

System design

Managing many autoresearch runs and extracting useful results requires dedicated engineering effort.

From research questions to formally verified results

The forward pass

- 1 Asking the right questions**
A mech interp researcher picks the macro question and turns it into a micro-question the agents can work on.
- 2 The harness**
Coordinates the agents that do the math and formally verify the results.
- 3 Mech interp dojo**
An open-source modular library in Lean 4 that formally verifies the mathematical results obtained during this project.

From research questions to formally verified results

The backprop

4 Inspecting the logs

Every autoresearch run is logged in Inspect, so it is easy to audit. This would inform how we can improve the system design.

5 Comparing harnesses

We benchmark harnesses on mathematical progress. That would inform us about effective collaboration strategies between models.

6 Steering autoresearch runs

We need to learn how to close the autoresearch loop for theoretical work, where progress cannot be reduced to a single machine-checkable metric.

Outlook

Summary

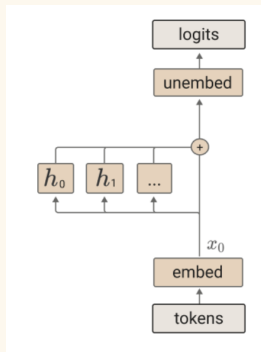
Theoretical mech interp is more accessible now

1 Theory can shape practice

Theoretical mech interp has the potential to shape empirical mech interp, as it has done before.

2 The barrier is lower now

With frontier models improving at mathematical reasoning and formal verification, problems in theoretical mech interp are within reach.



Elhage et al. (2021)

Summary

Vision: research taste and contribution

1 Problems that inform probes

We will work on problems whose answers inform theoretically grounded probes.

2 Tools we will share

Our harnesses and Lean codebase will be shared with mechanistic interpretability researchers, making their own autoresearch runs easier to maintain and audit.



`verifiedmechanisms.ai`

Team

Researchers in mechanistic interpretability and Lean formalisation

1

Who we are

An independent, grant-funded research project. The current team consists of researchers with a background in mech interp and Lean formalisation.

2

Who is joining

Thanks to grantmaking.ai, more researchers will join for the next 3 months to work on the pilot project, on the **mech interp** and **system design** aspects.

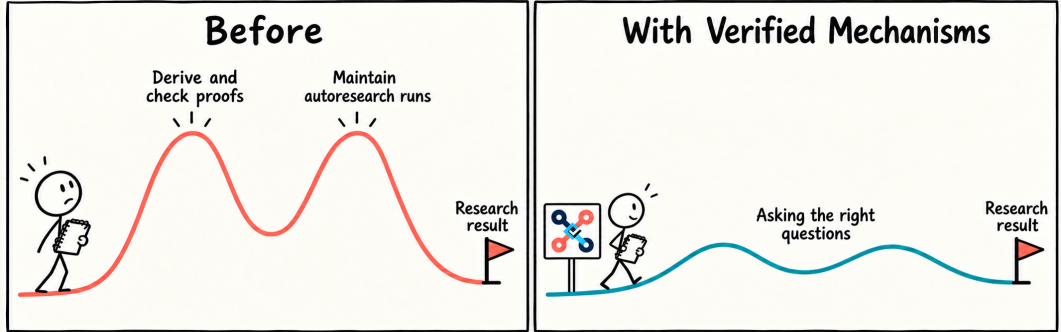
3

Supported by

grantmaking.ai and BlueDot Rapid Grant.

Verified Mechanisms, in pictures

Lowering the barriers to theoretical mech interp research.



Thank you

Questions and comments are welcome!

`verifiedmechanisms.ai` · `github.com/VerifiedMechanisms`

Supported by `grantmaking.ai` and BlueDot Rapid Grant



LW blog post



Discord



GitHub



MATS